

TUSHAR OJHA

B.Tech CSE Student | Python Developer | AI/ML, Data Analysis & Backend Systems
ojha55314@gmail.com | +91 9259725784 | Durgapur, WB / Roorkee, UK | github.com/to124578

PROFILE

Third-year Computer Science undergraduate who builds full, working systems rather than isolated exercises — spanning data pipelines, applied ML, backend APIs, and analytics dashboards. Backend lead on a 4-person hackathon team, published RAG researcher, and independent builder of production-style full-stack platforms. Comfortable across the stack: data cleaning and analysis, model training and evaluation, API design, and stakeholder-facing visualization.

TECHNICAL SKILLS

Languages: Python, Java (OOP), C, C++, JavaScript, TypeScript, SQL

AI/ML & Data: LangChain, LangGraph, RAG, XGBoost, scikit-learn, ChromaDB, pgvector, Ollama, Pandas, NumPy, Exploratory Data Analysis, Statistical Modeling

Backend & Frontend: FastAPI, Flask, Django, SQLAlchemy, REST APIs, React, TypeScript, Tailwind CSS, Leaflet.js

Tools & DevOps: Docker, GitHub Actions (CI/CD), pytest, Git, SQLite/PostgreSQL, Excel, Canva, basic IoT (Arduino)

PROJECTS

VayuDrishhti — AI-Powered Urban Air Quality Intelligence Platform

Solo Project, ET AI Hackathon 2026

- Built a full-stack platform end-to-end — data pipeline, XGBoost forecasting model, FastAPI/SQLAlchemy backend, and React/TypeScript/Leaflet frontend — forecasting AQI 24–72h ahead across 20 zones in Delhi NCR and Chennai.
- Achieved 70–74% lower forecast error (RMSE) than a naive baseline through iterative data cleaning, feature engineering, and model evaluation.
- Designed a geospatial dashboard with a ranked zone list and a what-if policy simulator, plus Claude API-powered multilingual advisories.
- Shipped with 32 passing pytest tests, GitHub Actions CI, and one-command Docker Compose deployment.

GuardianAI — Industrial Safety Intelligence Platform

Backend Lead, Team of 4, ET AI Hackathon 2026

- Led backend architecture for an AI platform flagging real-time safety risks from sensor/event data, using LangGraph multi-agent orchestration and a pgvector-backed RAG pipeline.
- Defined Pydantic event schemas as the data contract across agents; coordinated a 4-person cross-functional team under deadline.

Cherry — Voice-First Personal AI Assistant

Solo Project

- Built a FRIDAY-inspired voice assistant with multi-agent task distribution (Python, LangChain, Claude API) and a ChromaDB + SQLite memory layer, shipped as a 120+ file codebase with a React/Tauri desktop UI.

Published Research: Iterative RAG Framework

IJSCI, June 2026

- Published a paper proposing a custom evaluation metric (Retrieval Gap Coverage Score) for RAG systems in the International Journal of Sciences and Innovation Engineering (ISSN 3049-0251, Crossref-indexed).

Also built: IoT Attendance & Intrusion Detection System (Arduino Uno); ChatApp, a WhatsApp-style chat platform (raw sockets, SQLite); MapExplorer (Leaflet + Flask); a Universal File Compressor & Converter; a Hospital Management System (Django); plus a Telegram bot, contact book manager, and encoder/decoder suite.

LEADERSHIP & INVOLVEMENT

Backend Lead on the GuardianAI hackathon team (4 members) and self-led IoT project lead; participated in multiple hackathons and technical events via Unstop (Amity University, NIT, IIT-hosted).

CERTIFICATIONS

Walmart Global Tech — Advanced Software Engineering Job Simulation (Forage, Jul 2026) • Tata — Data Visualisation (Forage, Jul 2026) • HackerRank: Python, SQL, Software Engineer

EDUCATION

B.Tech in Computer Science and Engineering, Quantum University, Roorkee | 2024–2028 (Expected) | CGPA: 7.1/10